Lecture 9. Random Number Generation

Congruential generators, inverse CDF, Box–Muller, acceptance–rejection, multivariate sampling

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9.1 Uniform random numbers

All random number generation starts from i.i.d. Unif$(0,1)$ variates. Computers produce none: they produce deterministic **pseudo-random** sequences that only look random.

Multiplicative congruential generators. Take a large prime modulus m and a multiplier $a \in \{2,\dots,m-1\}$, and iterate $x_{k+1} = ax_k \bmod m$, outputting $u_k = x_k/m$.

Proposition 9.1

The map $f(x) = ax \bmod m$ is a bijection of $\{1,\dots,m-1\}$. The sequence is periodic with period $n = \min\{k \geq 1 : a^k \equiv 1 \bmod m\}$, which divides $m-1$.

Proof. $\gcd(a,m) = 1$ because m is prime, so ax is not a multiple of m for $1 \leq x \leq m-1$: f maps the set into itself. If $ax \equiv ay$, then $m \mid a(x-y)$, hence $m \mid x-y$ and $x = y$: f is injective, hence bijective on a finite set. Since $x_k = a^k x_0 \bmod m$, the sequence returns to x_0 exactly when $a^k \equiv 1$. By Fermat's little theorem $a^{m-1} \equiv 1 \bmod m$, so the period is at most $m-1$ (and divides it). ■

x_0 is the **seed**. If a is a **primitive root** modulo m (period $m-1$), the sequence runs through all of $\{1,\dots,m-1\}$. A classical choice is $m = 2^{31}-1$ (a Mersenne prime) and $a = 7^5 = 16807$, with period $2^{31}-2$ (used in early MATLAB).

Mixed congruential generators. $x_{i+1} = (ax_i + c) \bmod m$ with $c \neq 0$ allows $m = 2^M$, so that reduction modulo m is free in binary arithmetic. By the Hull–Dobell theorem the period is the full m iff $\gcd(c,m) = 1$, every prime factor of m divides $a-1$, and $4 \mid a-1$ when $4 \mid m$. Both $m = 2048$, $a = 1229$, $c = 1$ and the old C-library choice $m = 2^{32}$, $a = 1103515245$, $c = 12345$ satisfy this. But the first generator can produce only 2048 distinct values, multiples of $1/2048$, and repeats after 2048 draws, which is useless for simulation. The second has a long period, yet its low-order bits are poor: modulo 2^k the sequence has period only 2^k, so the last bit simply alternates. Plotting consecutive pairs (u_i, u_{i+1}) exposes the lattice structure of both.

A good generator needs a long period and output that passes statistical tests of uniformity and independence. Congruential generators have a structural flaw: consecutive p-tuples $(u_k,\dots,u_{k+p-1})$ lie on a small number of parallel hyperplanes in $[0,1]^p$ (Marsaglia's "random numbers fall mainly in the planes"; the notorious RANDU generator puts all triples on 15 planes). Shuffling helps. R's default, the **Mersenne Twister** (Matsumoto–Nishimura, 1998), has period $2^{19937}-1$ and is equidistributed in up to 623 dimensions.

9.2 The inverse CDF method

Motivation. If U is uniform, $F^{-1}(U)$ has cdf F. Whenever the quantile function is available in closed form, this is the simplest way to simulate from F.

For a cdf F define the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, $0 < u < 1$.

Theorem 9.2

Let $X \sim F$ and $U \sim$ Unif$(0,1)$.

(a) $F^{-1}(U)$ has cdf F.

(b) If F is continuous, $F(X) \sim$ Unif$(0,1)$.

(c) In general, $P\{F(X) \leq y\} \leq y$ for $y \in [0,1]$.

Proof. (a) We show $\{F^{-1}(u) \le t\} = \{u \le F(t)\}$. The set $\{x : F(x) \ge u\}$ is an interval $[F^{-1}(u), \infty)$ that contains its left endpoint, by monotonicity and right continuity; so $F(F^{-1}(u)) \ge u$. If $F^{-1}(u) \le t$ then $u \le F(F^{-1}(u)) \le F(t)$. Conversely, if $u \le F(t)$ then t belongs to the set, so $F^{-1}(u) \le t$. Hence $P(F^{-1}(U) \le t) = P(U \le F(t)) = F(t)$.

(b) First, $P\{F(X) \le F(t)\} = F(t)$ for every t: the event equals $\{X \le t\} \cup \{X > t, F(X) = F(t)\}$, and the second part is a set where F is flat, which has probability 0. For $u \in (0,1)$, continuity gives t with $F(t) = u$, so $P\{F(X) \le u\} = u$.

(c) By (a), we may take $X = F^{-1}(U)$, and $F(F^{-1}(U)) \ge U$. Hence $\{F(X) \le y\} \subset \{U \le y\}$. ■

Example 9.1

- Exponential, $F(x) = 1 - e^{-\lambda x}$: $F^{-1}(u) = -\frac{1}{\lambda}\log(1-u)$, and since $1 - U \overset{d}{=} U$ one uses $-\frac{1}{\lambda}\log U$.
- Cauchy(μ, β): $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{x-\mu}{\beta}$, $F^{-1}(u) = \mu + \beta\tan(\pi u - \frac{\pi}{2})$.
- Discrete uniform on $\{1, \dots, k\}$: $F^{-1}(u) = \lceil ku \rceil$.
- Geometric(p) on $\{1, 2, \dots\}$: if $Y \sim \text{Exp}(\lambda)$ with $1 - e^{-\lambda} = p$, then $\lceil Y \rceil$ is geometric, so $X = \lceil \log U / \log(1-p) \rceil$.

When F^{-1} has no closed form but F (or only the density f) is computable, $F^{-1}(u)$ can still be found by solving $F(x) = u$ numerically for each u: bisection on a bracketing interval, or Newton's method $x \leftarrow x - \{F(x) - u\}/f(x)$ (Lecture 6), with $F(x) = \int_{-\infty}^{x} f$ evaluated by quadrature if necessary. This is exact up to the tolerance but costly per draw, and for heavy tails (e.g. t_4) the bracketing interval must be wide enough.

9.3 Normal random numbers

Φ^{-1} has no closed form (though accurate rational approximations exist).

Box–Muller. For independent $X, Y \sim N(0,1)$ in polar coordinates (R, Θ), the joint density is $\frac{1}{2\pi}e^{-r^2/2}r$, so $\Theta \sim \text{Unif}(0, 2\pi)$ independently of R, and $P(R^2 > s) = e^{-s/2}$, i.e. $R^2 \sim \text{Exp}$ with mean 2. Hence for independent $U, V \sim \text{Unif}(0,1)$,

$$R = \sqrt{-2\log U}, \quad \Theta = 2\pi V, \qquad (X, Y) = (R\cos\Theta, R\sin\Theta)$$

are independent standard normals.

Marsaglia's polar method. If (U, V) is uniform on the unit disk, written $(T\cos\Phi, T\sin\Phi)$, then $P(T^2 \le t, \Phi \le \phi) = \frac{\pi t \cdot \phi/(2\pi)}{\pi} = t\frac{\phi}{2\pi}$, so $T^2 \sim \text{Unif}(0,1)$ and $\Phi \sim \text{Unif}(0, 2\pi)$ independently. So $-2\log T^2$ plays the role of R^2, and $(U/T, V/T)$ that of $(\cos\Theta, \sin\Theta)$:

(1) draw U, V i.i.d. Unif$(-1, 1)$;

(2) if $T^2 = U^2 + V^2 \ge 1$, return to (1);

(3) output $X = U\sqrt{-2\log T^2/T^2}$ and $Y = V\sqrt{-2\log T^2/T^2}$.

It avoids trigonometric functions at the price of using $4/\pi$ times as many pairs (the rejection step is an instance of Section 9.5).

9.4 Generating from definitions

- Bernoulli(p): $\mathbf{1}\{U \le p\}$. Binomial(n, p): sum of n Bernoullis. Negative binomial(r, p): sum of r geometrics.
- Poisson(λ): the number N with $\prod_{i=1}^{N} U_i \ge e^{-\lambda} > \prod_{i=1}^{N+1} U_i$. Indeed $T_i = -\log U_i/\lambda$ are i.i.d. Exp(λ) interarrival times, and N is the number of arrivals of a rate-λ Poisson process in $[0, 1]$: $T_1 + \cdots + T_N \le 1 < T_1 + \cdots + T_{N+1}$.
- $\chi^2(\nu)$: $\sum_{i \le \nu} Z_i^2$; for even ν, also $-2 \log \prod_{i \le \nu/2} U_i$, since $\chi^2(\nu) = \text{Gamma}(\nu/2, 2)$.
- $t(\nu) = Z/\sqrt{X/\nu}$ and $F(\nu_1, \nu_2) = (X_1/\nu_1)/(X_2/\nu_2)$ from independent normal and chi-square variates.

9.5 Acceptance–rejection sampling

Motivation. When F^{-1} is not available but the density is known up to a constant, we can sample from an easier envelope density and accept each draw with probability proportional to the ratio; the accepted draws have exactly the target distribution.

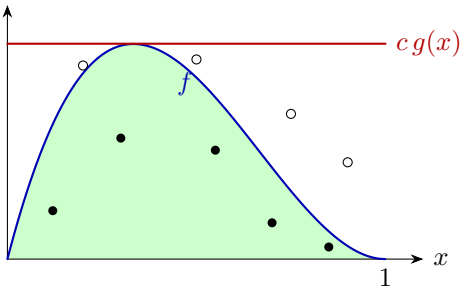


Figure. *Acceptance–rejection with target f (a Beta(2, 3) density) and uniform proposal g, $c = \sup f/g$: points uniform under $c\,g$ are kept if they fall under f (filled) and rejected otherwise (open). Accepted x-coordinates have density f; the acceptance rate is $1/c$.*

To sample from a density f with no tractable cdf (von Neumann):

(1) find a density g that is easy to sample and a constant c with $f \le cg =: h$ (an *envelope*);
(2) draw $X \sim g$ and an independent $U \sim \text{Unif}(0, 1)$;
(3) accept X if $U \le f(X)/h(X)$; otherwise return to (2).

The **body** of a nonnegative function f is $B_f = \{(x, y) : 0 \le y \le f(x)\}$, with area $|B_f| = \int f$.

Theorem 9.3

If $X \sim g$ and $U \sim \text{Unif}(0, 1)$ are independent, then $(X, cg(X)U)$ is uniform on B_{cg}.

Proof. Given $X = x$, $Y = cg(x)U$ is uniform on $[0, cg(x)]$, with density $\frac{1}{cg(x)}\mathbf{1}\{0 \le y \le cg(x)\}$. Multiplying by $g(x)$, the joint density is $\frac{1}{c}\mathbf{1}_{B_{cg}}(x, y)$; and $|B_{cg}| = c$. ■

Corollary 9.4

An accepted X has density f, and each proposal is accepted with probability $1/c$.

Proof. Acceptance is the event $(X, Y) \in B_f \subset B_{cg}$. Conditioning a uniform distribution on a subset gives the uniform distribution on the subset, so the accepted (X, Y) is uniform on B_f, with density $\mathbf{1}_{B_f}/|B_f|$. Its x-marginal is $\int_0^{f(x)} dy/|B_f| = f(x)/\int f = f(x)$. The acceptance probability is $|B_f|/|B_{cg}| = 1/c$. ■

So the envelope should be tight (c close to 1); the number of proposals per accepted draw is geometric with mean c. f need only be known up to a constant, as long as the envelope uses the same constant.

Example 9.2 (Uniform on the disk)

In Marsaglia's method, $f = \frac{1}{\pi}\mathbf{1}\{u^2 + v^2 < 1\}$ and $g = \frac{1}{4}\mathbf{1}_{[-1,1]^2}$, so $c = 4/\pi$ and $f/(cg)$ is the indicator of the disk: accept iff $U^2 + V^2 < 1$.

Example 9.3 (Beta variates from uniforms)

For Beta(a, b) with $a, b > 1$ the density is bounded by its value $d_{\max}$ at the mode $(a-1)/(a+b-2)$. Taking $g = \text{Unif}(0, 1)$ and $c = d_{\max}$: draw $Y \sim \text{Unif}(0, 1)$ and $U \sim \text{Unif}(0, 1)$, and keep Y iff $U \le f(Y)/d_{\max}$. The kept values are a Beta(a, b) sample and the acceptance rate is $1/d_{\max}$. Running this on 1000 proposals returns a random number (about $1000/d_{\max}$) of Beta draws. The method is simple but wasteful when the density is peaked (large a, b).

Example 9.4 (Gamma variates)

Since $c\,\text{Gamma}(\alpha, 1) = \text{Gamma}(\alpha, c)$ and $\text{Gamma}(\alpha, 1) \overset{d}{=} \text{Gamma}(\lfloor\alpha\rfloor, 1) \oplus \text{Gamma}(\alpha - \lfloor\alpha\rfloor, 1)$, where the first is a sum of exponentials, it suffices to handle $0 < \alpha < 1$. Then $x^{\alpha-1}e^{-x} \le x^{\alpha-1}$ on $[0, 1]$ and $\le e^{-x}$ on $(1, \infty)$, giving the envelope $h(x) = \{x^{\alpha-1}\mathbf{1}_{[0,1]} + e^{-x}\mathbf{1}_{(1,\infty)}\}/\Gamma(\alpha)$ with $c = (\frac{1}{\alpha} + \frac{1}{e})/\Gamma(\alpha)$. The proposal density $g = h/c$ has cdf $G(x) = x^\alpha/(1 + \alpha/e)$ on $[0, 1]$ and $1 - \alpha e^{-x}/(1 + \alpha/e)$ for $x > 1$, which inverts in closed form:

$$G^{-1}(u) = \begin{cases} \{(1 + \alpha/e)u\}^{1/\alpha}, & u \le 1/(1 + \alpha/e), \\ -\log\{(1/\alpha + 1/e)(1 - u)\}, & \text{otherwise.} \end{cases}$$

Example 9.5 (Gamma with $\alpha > 1$: a Cauchy envelope)

Let $\tilde{f}(x) = x^{\alpha-1}e^{-x}$ $(x > 0)$, $\alpha > 1$, $\mu = \alpha - 1$, $\sigma^2 = 2\alpha - 1$, and

$$g(x) = \frac{\mu^\mu e^{-\mu}}{1 + (x - \mu)^2/\sigma^2},$$

a multiple of the Cauchy(μ, σ) density. *Domination*: $k(x) = x^{\alpha-1}e^{-x}\{\sigma^2 + (x - \mu)^2\}$ has derivative $-x^{\alpha-2}e^{-x}(x - \alpha)^2(x - \mu)$, which is ≥ 0 for $x < \mu$ and ≤ 0 for $x > \mu$, so $k(x) \le k(\mu) = \sigma^2\mu^\mu e^{-\mu}$; dividing by $\sigma^2 + (x - \mu)^2$ gives $\tilde{f} \le g$ on $(0, \infty)$. *Algorithm*: draw $Y = \mu + \sigma\tan\{\pi(V - \frac{1}{2})\}$ with $V \sim \text{Unif}(0, 1)$ (inverse cdf); reject if $Y \le 0$; otherwise accept if $U \le \tilde{f}(Y)/g(Y)$. *Efficiency*: since $\int g = \pi\sigma\mu^\mu e^{-\mu}$ and $\int \tilde{f} = \Gamma(\alpha)$, the acceptance probability is

$$\frac{\Gamma(\alpha)}{\pi\sqrt{2\alpha - 1}\,(\alpha - 1)^{\alpha-1}e^{-(\alpha-1)}} \approx 0.47,\ 0.50,\ 0.54,\ 0.56 \quad (\alpha = 1.5, 2, 5, 100),$$

increasing to $1/\sqrt{\pi} \approx 0.564$ as $\alpha \to \infty$, by Stirling's formula $\Gamma(\alpha) = \Gamma(\mu + 1) \sim \sqrt{2\pi\mu}\,\mu^\mu e^{-\mu}$. So about two proposals per draw, uniformly in α, unlike the uniform envelope above.

9.6 Multivariate distributions

Batch methods.

- Multinomial$(n; p_1, \dots, p_k)$: draw n categories, category j when $U \in [\sum_{l<j} p_l, \sum_{l\le j} p_l)$, and count. (Alternatively, independent Poissons with means λp_j conditioned on their sum being n, which is inefficient.)
- Multivariate normal $N(\mu, \Sigma)$: find L with $LL^\top = \Sigma$ and set $\mu + LZ$, $Z \sim N(0, I_p)$. Use the Cholesky factor (cheapest) or the symmetric square root $Q\Lambda^{1/2}Q^\top$ from the eigendecomposition (which also handles singular Σ). For large p the factorization is the bottleneck.
- Multivariate $t_\nu(\mu, \Sigma)$: $\mu + X/\sqrt{W/\nu}$ with $X \sim N(0, \Sigma)$ independent of $W \sim \chi^2(\nu)$.

Sequential (conditional) sampling. Draw components one at a time from their conditional distributions:

- Multinomial: $X_1 \sim B(n, p_1)$, then $X_2 \mid X_1 = x_1 \sim B(n - x_1, p_2/(1 - p_1))$, and so on.
- Multivariate normal: $X_1 \sim N(\mu_1, \sigma_{11})$, then $(X_2, \dots, X_p) \mid X_1 = x_1 \sim N\big(\mu_2 + \Sigma_{21}(x_1 - \mu_1)/\sigma_{11},\ \Sigma_{22} - \Sigma_{21}\Sigma_{12}/\sigma_{11}\big)$, and recurse.

9.7 Selected exercises

Motivation. The first exercise proves the weakness of the low-order bits of power-of-two congruential generators mentioned above; the second designs an optimal acceptance–rejection sampler for the normal distribution.

Exercise 9.1 (Low-order bits of a mixed congruential generator)

Let $x_{i+1} = (ax_i + c) \bmod 2^M$ and, for $1 \le k \le M$, $y_i = x_i \bmod 2^k$ (the k lowest bits of x_i).
(a) Show that $y_{i+1} = (ay_i + c) \bmod 2^k$, and conclude that (y_i) is eventually periodic with period at most 2^k.
(b) Show that if a and c are odd, the last bit of x_i alternates between 0 and 1.

Solution. [Supplement]

(a) $x_{i+1} = ax_i + c - q2^M$ for an integer q, and 2^k divides 2^M, so $x_{i+1} \equiv ax_i + c \pmod{2^k}$. Since $x_i \equiv y_i \pmod{2^k}$, also $ax_i + c \equiv ay_i + c$, hence $y_{i+1} = (ay_i + c) \bmod 2^k$. So y_{i+1} is a fixed function of y_i with at most 2^k possible values. Among $y_0, \dots, y_{2^k}$ two coincide, say $y_s = y_{s+d}$ with $1 \le d \le 2^k$, and then $y_{i+d} = y_i$ for all $i \ge s$ by induction.

(b) For $k = 1$: a odd gives $ay_i \equiv y_i$ and c odd gives $c \equiv 1 \pmod 2$, so $y_{i+1} = (y_i + 1) \bmod 2$.

So in the generator $m = 2^{32}$, $a = 1103515245$, $c = 12345$ (both odd), the last bit alternates and the k last bits repeat with period at most 2^k, even though the full period is 2^{32}. Only the high-order bits should be used, e.g. $u_i = x_i/2^M$ and never $x_i \bmod 2$ as a coin flip. ■

Exercise 9.2 (Normal variates from a Laplace envelope)

Let ϕ be the $N(0, 1)$ density and $g_\lambda(x) = \frac{\lambda}{2}e^{-\lambda|x|}$ the Laplace density, $\lambda > 0$.
(a) Find the smallest $c(\lambda)$ with $\phi \le c(\lambda) g_\lambda$.
(b) Find the λ that maximizes the acceptance probability, and that probability.
(c) Show that for the optimal λ the sampler can be run with exponential variates only.

Solution. [Supplement]

(a) The ratio $\phi(x)/g_\lambda(x) = \frac{2}{\lambda\sqrt{2\pi}}\exp(-\frac{x^2}{2} + \lambda|x|)$ depends on $|x|$, and the exponent $-\frac{t^2}{2} + \lambda t$ is maximized over $t \geq 0$ at $t = \lambda$, with value $\lambda^2/2$. So

$$c(\lambda) = \sup_x \frac{\phi(x)}{g_\lambda(x)} = \sqrt{\frac{2}{\pi}}\,\frac{e^{\lambda^2/2}}{\lambda}.$$

(b) The acceptance probability is $1/c(\lambda)$ (Corollary 9.4), so we minimize $\log c(\lambda) = \text{const} + \frac{\lambda^2}{2} - \log\lambda$. Its derivative $\lambda - 1/\lambda$ vanishes only at $\lambda = 1$, and its second derivative $1 + \lambda^{-2} > 0$, so $\lambda = 1$ is the global minimizer on $(0, \infty)$. Then $c(1) = \sqrt{2e/\pi} \approx 1.3155$ and the acceptance probability is ≈ 0.760.

(c) With $\lambda = 1$, $\phi(x)/\{c(1)g_1(x)\} = \exp\{-\frac{x^2}{2} + |x| - \frac{1}{2}\} = \exp\{-\frac{1}{2}(|x| - 1)^2\}$. A Laplace variate is $X = SE_1$ with $E_1 \sim \text{Exp}(1)$ and an independent random sign S, and the test $U \leq e^{-(E_1-1)^2/2}$ is equivalent to $E_2 \geq (E_1-1)^2/2$ with $E_2 = -\log U \sim \text{Exp}(1)$. So: draw independent $E_1, E_2 \sim \text{Exp}(1)$ until $E_2 \geq (E_1 - 1)^2/2$, then return $\pm E_1$ with a fair random sign. On average 1.3155 pairs are needed per normal variate. ■